Delegated execution

Own the job and its outbox. Let other owners commit their effects.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A service desk with a job ticket: “accepted,” “finished” and “customer notified” have separate receipts.

Why it matters: Delegated work crosses storage and messaging boundaries. Explicit jobs and an outbox allow recovery without a transaction spanning all three owners.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

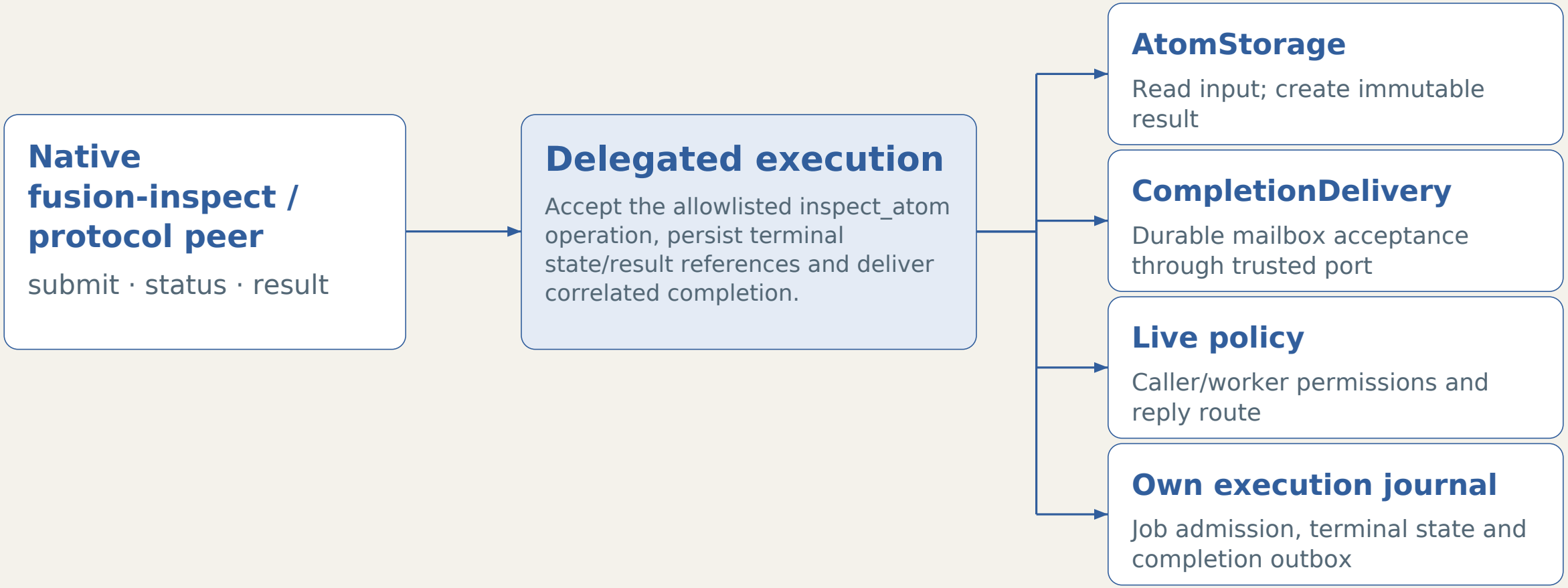
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	**Execution**
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Accept the allowlisted inspect_atom operation, persist terminal state/result references and deliver correlated completion.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Job + terminal decision

Queued/succeeded/failed and result/error correlation

Completion outbox

Delivery intent and recorded mailbox-acceptance progress

OUTSIDE THIS RESPONSIBILITY

Storage or mailbox mutation authority

Calls declared ports; no foreign tables or cross-owner SQL transaction

Reasoning/process migration

Only inspect_atom; no arbitrary code execution or cognition

inspect-atom/2 · operation algorithm 0.1.0 · public wire 1.0.0

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
submit	Validate caller/worker read permissions and input	Durable queued acceptance, not completion
status	Read caller-owned job and delivery observations	Execution and delivery remain distinct
result	Return accepted result reference after success	Reauthorize read; failed/pending outcomes explicit
Trusted worker / flush	Compute bounded result, persist and deliver	Not exposed as arbitrary client commands

One bounded worker writer; finite configured queued-job + unsent-outbox capacity.

A worked example through the boundary

ACTUAL native Hive inspect + real DeTTa · deterministic worker; fixture counterparty

01 · STEP

Accept input

Read the generated atom and accept inspect_atom.

02 · STEP

Persist result

Store digest, byte length and algorithm version.

03 · STEP

Commit terminal

Save result reference + outbox in one owner commit.

04 · STEP

Deliver / recover

Mailbox accepts; caller verifies and acknowledges.

What the example establishes

Execution completion and completion delivery have different owners.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Allowlisted work

Only the named operation is accepted; no caller-supplied code.

I2

Result before terminal

A terminal success references already persisted result bytes.

I3

Atomic outbox intent

Terminal state and completion intent commit together.

I4

Live authority/deadline

Restart rechecks caller/worker permissions; no new work after start deadline.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Result persisted; terminal commit lost	Retry immutable result creation, then commit terminal/outbox.	Stable result identity
Mailbox accepted; sent flag lost	Retry handoff through CompletionDelivery.	Mailbox owns deduplication
Unavailable input / revoked worker	Keep recoverable work or record explicit terminal failure as specified.	No unauthorized success

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

1 / 1

Job / outbox in preserved workflow

Stable across recovery matrix

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Generated M4 result payload

Exact accepted artifact, not a throughput figure

13

Original native inspect cases

Preserved M1.2 suite; not 13 unique new M5 cases

The fixed inspect operation demonstrates orchestration and recovery, not compute efficiency. No standalone job-admission latency, worker throughput or fairness benchmark is recorded.

PROPOSED BENCHMARK · NOT RUN

Proposed: vary input size, queue depth and downstream failure; measure admission/result/notification separately, including duplicate physical attempts and pending outbox age.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Local persistent job/outbox, separate worker process, actual native Hive and real DeTTa.
FIXTURE / LOCAL TEST POLICY	Generated inspect workload, protocol counterparty, local grants and deterministic fault placement.
BLOCKED / UNSUPPORTED	No required current gate blocked; unavailable input or full/revoked route may leave progress pending.
UNRUN	Arbitrary workloads, remote worker scheduling, distributed failover and exactly-once physical execution.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Benchmark each lifecycle stage	Independent timestamps for acceptance/result/delivery	Do not report notification delay as execution time
Add an operation explicitly	Determinism, resource bounds and authorization review	New conformance and crash boundary tests
Explore remote workers	Placement and failure contract before transport	Keep independent owner commits and stable replay

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Execution completion and completion delivery have different owners.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m11.py
```

For actual Hive, use the pinned M1.2 build/run recipe or the current full legacy verifier; the M1.1 command alone is not a native-caller gate.

`fusion/execution.py`
`fusion/service.py`
`tests/test_m11.py`
`integration/hive/test_integration.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.